

Demonstration of Proposed Features

Date	28/06/2026
Team ID	Not Assigned
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Exploratory Data Analysis (EDA)	Univariate (distribution plots, box plots), multivariate analysis (correlation heatmap, violin plots) and descriptive statistics on the flood dataset.	Implemented	Yes	Fully completed in flood_eda.ipynb
2	XGBoost Flood Prediction Model	Trained XGBoost classifier on 10 meteorological features; StandardScaler applied for preprocessing. Achieves 96.55% accuracy.	Implemented	Yes	Model saved as floods.save; scaler as transform.save
3	Flask Web Application	4-route Flask app (/ , /Predict, /chance, /no_chance) serving Jinja2 templates. Accepts 10 weather inputs and returns flood risk.	Implemented	Yes	app.py — all routes functional
4	Input Form & Validation	HTML form collects Temp, Humidity, Cloud Cover, ANNUAL, Jan-Feb, Mar-May, Jun-Sep, Oct-Dec, avgjune, sub. Server-side validation with error handling.	Implemented	Yes	index.html + app.py error handling
5	Result Pages (Flood / No Flood)	Displays flood probability (%) and input summary on chance.html or no_chance.html based on XGBoost prediction.	Implemented	Yes	chance.html & no_chance.html rendered correctly
6	Cloud Deployment (Render.com)	Application deployed publicly via Unicorn on Render.com. Accessible globally through a live URL.	Implemented	Yes	Live at rising-waters-flood-prediction.onrender.com

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%